

Microfinance Loan App for SHGs

Domain: Digital Lending | Statutory Architecture

Parameter	Evaluation Result	Governance & Statutory Impact
MoSCoW Baseline	Must Have	Ensures compliance with regulatory guidelines and enhances security measures.
Risk Classification	HIGH-IMPACT / CRITICAL (92/100)	High financial asset impact; requires automated verification gates
Total Requirements	8 Modular Requirements	Decomposed into Frontend, Backend, Auth, Security, Database, and Resilience

1. System Scope & Stakeholder Requirement Statement

"Build a microfinance loan app for rural self-help groups (SHG) with Aadhaar-based biometric consent and automated loan disbursement via DBT."
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2. Modular Software Requirements Specification (8 Detailed Points)

Pillar: Integration (1 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-01	Automated Loan Disbursement via Direct Benefit Transfer (DBT) The system will automatically transfer loan amounts from the government database to the borrower's account via DBT.	Must Have	Regulation: 1724326790365.pdf Acceptance: Loan amount transferred successfully to the borrower's account within 24 hours.

Pillar: Security (2 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-02	Biometric Authentication for User Identification Users will authenticate their identity using biometric data such as fingerprints or iris scans provided by the Aadhaar card.	Must Have	Regulation: application_form_guidelines_for_submission_of_proposals_for_use_of_aadhaar_authentication.pdf Acceptance: Authentication successful with no false negatives or positives within 3 seconds.
REQ-05	Secure Data Storage and Transmission Data related to users, transactions, and loan information is stored securely on servers and transmitted over encrypted channels.	Must Have	Regulation: Statutory Compliance Directive Acceptance: No unauthorized access or data breaches reported within one year.

Pillar: Interface (1 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-03	Rural Financial Transactions Processing The app supports transactions related to rural financial activities including loan applications, repayments, and other financial operations.	Should Have	Regulation: Statutory Compliance Directive Acceptance: Supports all specified financial transactions without errors within 5 minutes.

Pillar: Resilience (1 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-04	Compliance with Government Regulations The system adheres to all applicable government regulations and policies regarding financial transactions and user privacy.	Must Have	Regulation: 1724326790365.pdf Acceptance: <i>Complies with all necessary regulations and receives regular audits.</i>

Pillar: Monitoring & Reporting (1 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-06	Real-time Transaction Monitoring The system continuously monitors real-time transaction details and generates reports for management review.	Should Have	Regulation: Statutory Compliance Directive Acceptance: <i>Generates detailed reports every hour with accuracy up to 99%.</i>

Pillar: User Experience (1 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-07	User Feedback Mechanism Users can provide feedback on their experience which will be used to improve the service quality.	Nice to Have	Regulation: Statutory Compliance Directive Acceptance: <i>At least 90% user satisfaction rate after three months.</i>

Pillar: Localization (1 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-08	Multi-Language Support The app supports multiple languages to cater to different regional preferences and needs.	Should Have	Regulation: Statutory Compliance Directive Acceptance: <i>Available in at least five major Indian languages.</i>

3. Regulatory & Statutory Grounding (Indian Banking Frameworks)

Framework	Enforcement Level	Statutory Directive / Citation
1724326790365.pdf	Mandatory Compliance	1724326790365.pdf Directives
application_form_guidelines_for_submission_of_proposals_for_use_of_aadhaar_authentication.pdf	Mandatory Compliance	application_form_guidelines_for_submission_of_proposals_for_use_of_aadhaar_authentication.pdf Directives

4. Mandatory Security & Architectural Controls

Control ID	Title & Architecture Specification	Automated Verification Method
CTRL-01	TLS 1.3 Transport Security & End-to-End Encryption All client-server communications and API calls for Microfinance Loan App for SHGs enforce TLS 1.3.	SSL/TLS automated protocol analyzer & penetration test.
CTRL-02	Immutable Forensic Audit Logging Append-only, tamper-evident audit logging for all critical operations.	Log integrity checksum validation & WORM verification.

5. Traceability Matrix & Recommended SDLC Sprints

Req IDs	Target Sub-Module	SDLC Phase / Gate	Verification Deliverable
REQ-01	Microfinance Loan App for SHGs — Integration	Sprint 1 / Integration Milestone	Integration Module & Verification Test Suite <i>Automated Unit/Integration Tests & Security Signoff</i>
REQ-02 - REQ-05	Microfinance Loan App for SHGs — Security	Sprint 2 / Security Milestone	Security Module & Verification Test Suite <i>Automated Unit/Integration Tests & Security Signoff</i>
REQ-03	Microfinance Loan App for SHGs — Interface	Sprint 3 / Interface Milestone	Interface Module & Verification Test Suite <i>Automated Unit/Integration Tests & Security Signoff</i>
REQ-04	Microfinance Loan App for SHGs — Resilience	Sprint 4 / Resilience Milestone	Resilience Module & Verification Test Suite <i>Automated Unit/Integration Tests & Security Signoff</i>
REQ-06	Microfinance Loan App for SHGs — Monitoring & Reporting	Sprint 5 / Monitoring Milestone	Monitoring & Reporting Module & Verification Test Suite <i>Automated Unit/Integration Tests & Security Signoff</i>
REQ-07	Microfinance Loan App for SHGs — User Experience	Sprint 6 / User Milestone	User Experience Module & Verification Test Suite <i>Automated Unit/Integration Tests & Security Signoff</i>
REQ-08	Microfinance Loan App for SHGs — Localization	Sprint 7 / Localization Milestone	Localization Module & Verification Test Suite <i>Automated Unit/Integration Tests & Security Signoff</i>